

Heuristic Method of Teaching Mathematics

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Heuristic Method / Discovery Method

Introduction

- Method of discovery
- Method of independent discovery pursued by discoverers and inventors
- Keeps the students as far as possible in the attitude of a discoverer
- Derived from the Greek word 'heurisco', which mean 'I find'.
- Involves finding out by the students instead of merely telling of everything by the teacher
- The teacher's job is not to solve the problems for the pupils, but to enable the pupils to solve problems for himself

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Origin

- Prof. Henry Edward Armstrong (British Chemist) was the originator of this method
 - unsatisfied with the state of affairs of teaching science
 - science could not be taught or learned through a textbook
 - scientific information and facts should be gathered by the students with their own independent effort

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Features

- Develop the thinking power of the students, and for this, they must be given an opportunity to think
- puts the child in a position of thinker and developer
- aims at training learners in the methods of learning. Self-confidence, originality, power of thinking and independent judgement are to be developed in individual to make him as ever successful students

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Features

- Heuristic method as the name implies is a method by which the pupils discover things for themselves. The pupil is put in the position of a pioneer, and they find their way along the path of knowledge as did those who first discovered the facts, principles and laws which are now known to all.

- **Rybur W. M.**

- Essentially, the Heuristic Method is intended to provide training in method and knowledge is a secondary consideration altogether.

- **Westway**

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Procedure to be adopted in using Heuristic Approach

Planning

- Deterring Aims and Formulating Objectives
- Selection of the Problems
- Framing a tentative Solution to the Problem

Execution

- Investigate / Discover / Observe in search of solution
- Infer / Interpret Observation

Conclusion

- Draw Conclusion and generalise
- Arrive at Solution / new Knowledge

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Procedure to be adopted in using Heuristic Approach

In adopting the heuristic method, all the children in a class may be asked to work at the same problem simultaneously, and each child may be made to feel responsible for finding out something for themselves. Each one makes independent inquiries; freedom of movement, discussion and questioning is allowed in the classroom and laboratory. The essence of this method involves questioning and students try to find out the answers to such questions as far as possible by themselves and if necessary, consult the teacher.

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Procedure to be adopted in using Heuristic Approach

Each child tries to acquire information about the problem from different sources. They try to move about and discuss the problem with their classmates. Each student is given a sheet of instructions regarding the problem in hand. The student can get guidance from the teacher. The teacher should try to induce answers from students by raising as many questions as possible. In this way, they interpret their observations and finally draw out conclusions.

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The role of the teacher in teaching through this method

1. The teacher should be resourceful and cultivate the heuristic spirit in himself. He needs special preparation for this.
2. Through well-choosen questions, the teacher leads pupils to discover facts and relationship for themselves from learning situations. Indeed this is the best procedure for introducing new steps in the teaching of mathematical topics. They learn by their own efforts, and therefore they are motivated to learn. The success of this method will be judged to the extent the teacher is able to handle.
3. The utmost sincerity, patience and alertness on the part of the teacher are called for.
4. The teacher should help students to identify a suitable problem. Here the teacher should keep in mind the age, ability and level of the students, along with the facilities available.

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The role of the teacher in teaching through this method

5. Overdoing this method for the purpose of eliciting silly or very obvious thing will make it a ridiculous tool. The teacher should take care to avoid the false question.
6. The teacher should act as a guide and teacher to the students providing them with sufficient background information and help the students to get going.
7. Detailed instruction sheet should be provided by the teacher to students.
8. Arrangements for supplementing the students' knowledge should be made by teacher be it library facilities or field trip.
9. The teacher should possess curiosity, interest and spirit of scientific investigation because they are to be developed in the child. Unless the teacher is a discoverer, one can not expect the children under their guidance to develop the attitude of a discoverer.

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The role of the teacher in teaching through this method

10. This method demands a lot of questioning. The teacher should be adept in the art of questioning, which encourages mental activity in the children, and the teacher should also encourage the students to ask a question.
11. The teacher should ensure an atmosphere of freedom so that children are encouraged towards self-development, spontaneity and self-expression.

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Utility of Heuristic Methods of Teaching

1. Home study and cramming are substantially reduced.
2. It imparts knowledge of the scientific method of thinking and inculcates scientific attitude among students.
3. It gives children happiness, sense of achievement, mental satisfaction and encouragement when the child successfully discovers something by their efforts.
4. The method lays stress on individual practical work, careful observation and independent thinking, which makes a child self-reliant.
5. It suits the subject because it involves thinking, reasoning and discovery.
6. The teacher-taught contents are very intimate.

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Utility of Heuristic Methods of Teaching

7. The child acquires a real understanding and a clear notion of the subject.
8. It suits the child. He becomes an active participant in the process of learning.
9. It is a psychologically sound method as it aims to utilize the active, original and constructive tendencies of the learner.
10. The individual difference can be catered to, as each student can work at their own pace.
11. The child becomes a thinker and discoverer instead of becoming a crammer.
12. By allowing students, freedom of thought and action will enable them to develop value and attitude consistent with the nature of Mathematics.

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Utility of Heuristic Methods of Teaching

13. By allowing students, freedom of thought and action will enable them to develop value and attitude consistent with the nature of Mathematics.
14. Students develop research skill, including setting up and fitting apparatus, conducting experiments, the habit of neatness and patience are acquired.
15. It prepares a student for life; it gives them the ability to tackle and solve any problem of life scientifically.

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Limitation in Heuristic Method of Teaching

1. The immature child and an average teacher may not come up to the mark in this method.
2. There is very little hope that the child left to himself will show reasonable continuous progress.
3. It demands great labour and special preparation from the already overburden teacher.
4. In lower classes, especially, the child needs hints and guidance. If the teacher does not give him prompt guidance, he may get discouraged.
5. The teacher has to give well-measured guidance, neither more nor less.
6. It is not equally applicable to all topics of the subject.
7. It demands individual attention which may not be possible in a large class.
8. If the students are tempted to consult books and copy them for the sake of discovery, the method fails.

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Conclusion

The use of an extreme form of this method is out of the question. The teacher's presence in the classroom must be purposeful. He is not to remain an indifferent onlooker, but his company is to inspire and stimulate the learners. In actual practice, the success of this method depends largely on good questioning. It is the spirit behind this method. The learner no longer listens but finds. The teacher should remember that he has to replace his teaching with discovery, thinking and learning on the part of the students.

This method is not applicable in primary classes but can be practicable in Middle and High school classes because by then students have gained enough knowledge of basic principles and procedures, it is difficult to be adopted for it requires resourceful teachers, infrastructure and time. But it is essential that a Heuristic atmosphere should exist in every class. There should be no spoon-feeding on the part of the teacher. The pupil should be made to think and work independently.



Discussion

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